

API System Reliability Guide

1. Overview

This document defines the API reliability, troubleshooting, escalation, and recovery procedures for FlowSync platform services.

It is used by:

- Support agents
- Engineering teams
- AI troubleshooting agents
- Incident management workflows

The goal is to ensure:

- Faster diagnosis
 - Consistent escalation handling
 - Reduced downtime impact
 - Intelligent automated support decisions
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2. Common API Failures

2.1 Timeout Errors

Description

Requests exceed expected response time limits.

Common Causes

- Backend overload
- High database latency
- Network congestion

- Third-party dependency delays
- Large payload processing

Symptoms

- API requests hanging
- Delayed dashboard loading
- Integration sync delays

Severity Mapping

- Single endpoint affected → P2
 - Multiple endpoints affected → P1
 - Platform-wide timeout → P0
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2.2 500 Internal Server Error

Description

Unexpected server-side processing failure.

Common Causes

- Service crash
- Unhandled exceptions
- Dependency failure
- Misconfigured environment variables
- Deployment instability

Symptoms

- Immediate request failure
- Broken workflows
- Failed integrations

Severity Mapping

- Isolated service → P2
- Core workflow affected → P1

- Authentication or billing affected → P0
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2.3 Rate Limit Exceeded

Description

Request threshold exceeded within defined time window.

Common Causes

- Too many requests per minute
- API abuse
- Misconfigured integrations
- Infinite retry loops
- Free plan request limitation

Symptoms

- HTTP 429 errors
- Integration sync interruptions
- Delayed webhook processing

Severity Mapping

- Single user impacted → P3
 - Enterprise integration affected → P1
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2.4 Authentication API Failure

Description

Authentication or authorization services failing.

Common Causes

- Token validation service outage
- OAuth provider issues

- Session management failure
- Expired signing certificates

Symptoms

- Login failures
- MFA verification errors
- Session expiration loops

Severity Mapping

- Limited users impacted → P1
 - Platform-wide auth outage → P0
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2.5 Database Connectivity Failure

Description

API services unable to communicate with database systems.

Common Causes

- Database downtime
- Connection pool exhaustion
- High query latency
- Network segmentation

Symptoms

- API timeouts
- Missing data
- Failed transactions

Severity Mapping

- Partial degradation → P1
 - Complete outage → P0
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3. Retry Strategy

Standard Retry Policy

- Retry failed requests 3 times
 - Use exponential backoff strategy
 - Increase timeout progressively
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Retry Timing Example

Attempt	Delay
Retry 1	2 seconds
Retry 2	5 seconds
Retry 3	10 seconds

Maximum timeout:

- 15 seconds per request
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Retry Restrictions

Do NOT retry:

- Authentication failures caused by invalid credentials
 - Permission-denied errors
 - Malformed request payloads
 - Deleted resource requests
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4. Escalation Rules

Escalate immediately when:

- API failure persists longer than 30 minutes
- Enterprise customer is impacted

- Multiple endpoints fail simultaneously
 - Authentication systems are unstable
 - Billing APIs are affected
 - Error rate exceeds critical threshold
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5. Incident Severity Classification

Severity	Description
P0	Platform-wide outage
P1	Major API degradation
P2	Partial endpoint failure
P3	Minor intermittent issue

6. Recommended Fixes

6.1 Initial Diagnostics

- Check API health endpoints
 - Verify database connectivity
 - Review application logs
 - Inspect deployment status
 - Validate third-party dependency status
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6.2 Infrastructure Actions

- Restart API gateway
 - Restart failed service containers
 - Scale backend instances
 - Flush queue backlog
 - Reconnect database pools
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6.3 Integration-Specific Checks

- Validate webhook delivery
 - Verify OAuth tokens
 - Check API rate limits
 - Inspect integration permissions
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7. Monitoring Indicators

The following metrics should be monitored continuously:

- API response latency
 - Error rate percentage
 - Request throughput
 - Queue processing delays
 - Authentication failure rate
 - Database query latency
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8. AI Decision Support Rules

Auto-Resolve Conditions

AI may auto-resolve when:

- Known retryable timeout occurs
 - Temporary integration reconnect fixes issue
 - Documentation-based fix available
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Human Escalation Conditions

AI must escalate when:

- Severity = P0 or P1
- Enterprise SLA risk detected
- Multiple recovery attempts fail

- System-wide outage suspected
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9. Automated Ticket Generation Rules

The AI system should automatically create tickets when:

- API failure persists beyond retry threshold
- Error rate exceeds safe limit
- Customer reports repeated failures
- Enterprise integrations are affected

Generated ticket must include:

- Error summary
 - Endpoint impacted
 - Severity level
 - Retry attempts
 - Logs/context (if available)
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10. AI Usage

This document is used by AI agents for:

- Technical diagnosis
 - Severity classification
 - Escalation decisions
 - Automated ticket generation
 - Infrastructure troubleshooting
 - Incident response coordination
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